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Department of Examinations

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Menara Perera

Author and Developer of StudySphere Mobile App

Abstraction

StudySphere is an innovative Android-based m-commerce application specifically designed to create a collaborative marketplace for the student community. The platform addresses the common challenges students face in sourcing affordable academic resources suchas textbooks, stationery, guidebooks, and high-quality study notes within their local campus environment. Built using **Java** and **Android Studio**, the applicationleverages **Firebase Cloud Firestore** for real-time data management and **SQLite** for local notification storage.

The buyer's journey begins with a dynamic category-based product exploration system. The **Discover** module enhances user experience by offering advanced filtering based on product ratings and geographical proximityusing the **Google Maps API**. Users canmanage a personal **Wishlist** or additems to a synchronized **Shopping Cart**, which features real-time price calculations and stock availability validation. The checkout process ensures high security by integrating **Biometric (Fingerprint) Authentication** before proceeding to the **PayHere** payment gateway. Post-payment, the system automatically generates a digital receipt saved to the device's **Internal Storage** and logs a payment notification into a local SQLite database for history tracking.

A unique feature of StudySphere is its **Role-Based Access Control**. Abuyer can transition into a seller via a secure **OTP Verification** process linked to their mobile number. The **Seller Module** empowers students to listproducts by capturing their **current GPSlocation** or searching for a specific pickuppoint on the map. Sellersare provided with a comprehensive dashboard to manage product listings, upload and edit images, monitor customer reviews, and track incoming orders. Furthermore, the application includes a profile management system allowing users to update personal details and profile imagery. By integrating location-based services withsecure financialtransactions, StudySphere provides a robust and sustainable ecosystem for peer-to-peer academic resource exchange.

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1. Introduction

In the modern academic landscape, students frequently encounter difficulties in accessing essential learning materials such as textbooks, stationery, specialized lab equipment, and high-quality study notes. While e-commerce platforms exist, they often lack a localized, student-centric approach, leading to high costs and logistical delays. **StudySphere** is developed as a comprehensive Android-based m-commerce solution designed to bridge this gap by facilitating a secure and efficient peer-to-peer marketplace within the student community.

The primary objective of StudySphere is to empower students to buy and sell academic resources easily. The application integrates advanced mobile technologies to ensure a seamless user experience. One of its standout features is the **Location-Based Service** powered by Google Maps API, which allows sellers to register specific pickup points and enables buyers to discover resources within their immediate vicinity. This localized approach minimizes shipping costs and encourages face-to-face interactions among peers.

Security and reliability are core pillars of the StudySphere ecosystem. The application employs a **Role-Based Access Control** mechanism, requiring **OTP (One-Time Password) verification** for users to transition from a Buyer to a Seller. Furthermore, the checkout process is fortified with **Biometric Authentication**, ensuring that financial transactions via the **PayHere** gateway are authorized only by the legitimate account holder.

Technologically, the app is built on a robust framework using **Java** and **Android Studio**, utilizing **Firebase Cloud Firestore** for real-time data synchronization and **SQLite** for localized data logging. With additional features like **automated PDFreceipt generation**, **real-time stock management**, and **intelligent product filtering**, StudySphere aims to transform how students manage their educational resources, fostering a more sustainable and collaborative learning environment.

1.1 Motivation: The main motivation behind this project is the rising cost of educational materials. Many students have valuable resources they no longer use, while others struggle to buy them new. StudySphere aims to create a sustainable cycle of resource sharing within campuses.

1.2 Problem Statement: Existing methods for trading student resources, such as social media groups or physical notice boards, are fragmented, insecure, and lack real -time updates. There is no dedicated platform that allows students to find items based on their exact campus location with verified security.

2. Background

The **StudySphere** project addresses the critical problem of inefficient and fragmented access to academic resources within the student community. Currently, students face significant hurdles in finding affordable textbooks, stationery, and study notes, often relying on disorganized social media groups or physical notice boards that lack security and real-time updates.

While several e-commerce platforms exist, they are primarily designed for commercial retailers and fail to cater to the unique, localized needs of peer-to-peer student transactions. Existing solutions do not provide integrated features such as campus-based proximity searching, verified student-to-seller transitions, or secure biometric-authorized payments tailored for academic exchanges. This gap in the current market demonstrates that the problem of resource accessibility for students has not yet been fully solved, necessitating a dedicated, location-sensitive mobile marketplace like StudySphere.

3. Specification

This section outlines the functional and non-functional requirements, system architecture, and constraints of the **StudySphere** platform. The specification serves as a blueprint for the system's functionality, guiding the development process and ensuring that all stakeholders have a shared understanding of the project's goals.

3.1 Functional Requirements

These requirements define the specific behaviors and feature the system must provide to the users.

- **User Authentication & Profiles:** Secure login/registration and the ability to manage user profiles, including profile picture updates.
- **Dual-Role Management:** A mechanism to switch from a Buyer to a Seller role via **Mobile OTP Verification**.
- **Product Management (Seller):** Tools for sellers to add, edit, and delete academic resources, including image uploads and stock management.
- **Geolocation Services:** Integration with **Google Maps API** for setting pickup locations (via current GPS or manual search) and calculating the distance between buyers and sellers.
- **Discovery & Filtering:** A "Discover" module for buyers to filter products based on ratings and proximity (nearest items).
- **Transaction Workflow:** A complete shopping cart system, wishlist functionality, and a secure checkout process.
- **Biometric Security:** Integration of **Fingerprint Authentication** during the payment process.
- **Payment & Documentation:** Integration with the **PayHere** gateway and automated **PDF Receipt Generation** saved to internal storage.
- **Notifications & History:** Real-time notification logging using **SQLite** for offline access to payment history.

3.2 Non- Functional Requirements

- These requirements define the quality attributes and operational constraints of the system.
- Security: High-level data protection using Firebase Security Rules and biometric encryption for financial transactions.
- Performance: Real-time data synchronization with Cloud Firestore to ensure minimal latency in product updates.
- Usability: A student-centric, intuitive UI/UX design developed in Android Studio for ease of navigation.
- Availability: Ensuring the marketplace is accessible 24/7 with reliable cloud-based data storage.
- Reliability: Accurate distance calculations and error handling during payment gateway failures.

3.3 Constraints

- **Platform Constraint:** The application is developed exclusively for the **Android operating system**.
- **Connectivity:** A stable internet connection is required for real-time database synchronization and Google Maps functionality.
- **Payment Gateway:** The transaction system is limited to the features and sandbox environment provided by the **PayHere API**.
- **Storage:** The generation and storage of PDF receipts are dependent on the available internal storage of the user's mobile device.

4. Design

System Architecture Diagram

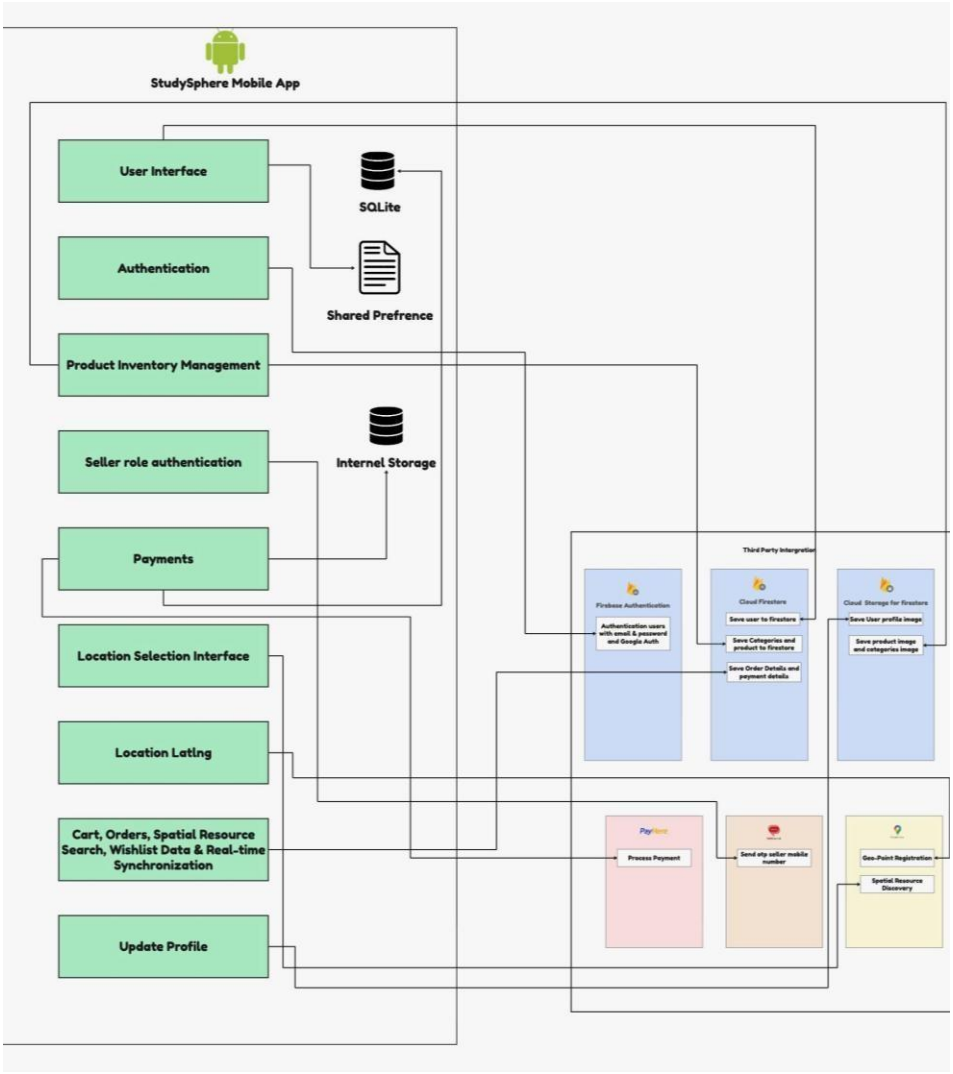


Figure 1: Studysphere System Architecture

Use Case Diagram for Buyer

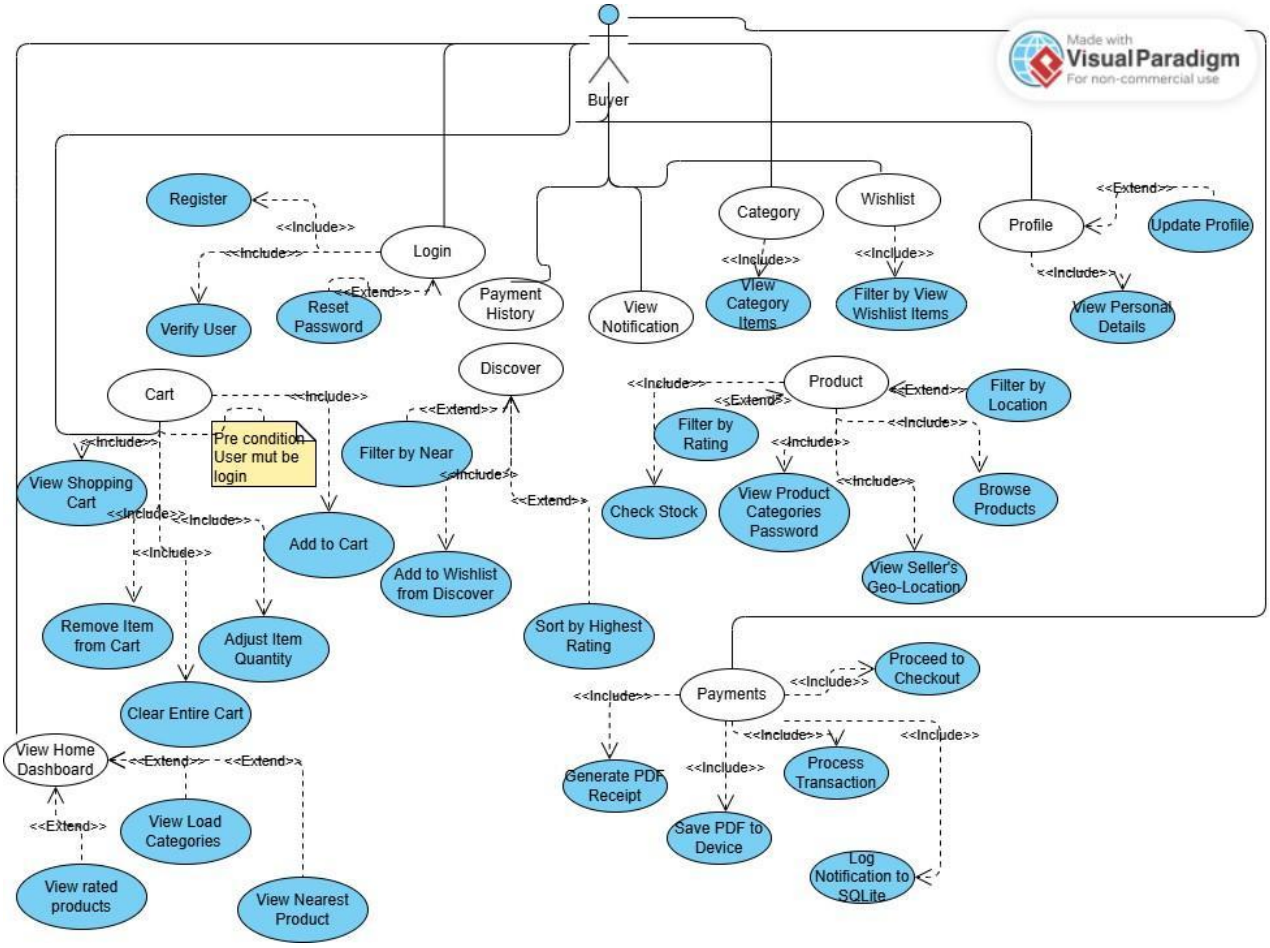


Figure 2: Studysphere Use Case Diagram (buyer)

Use Case Diagram for Seller

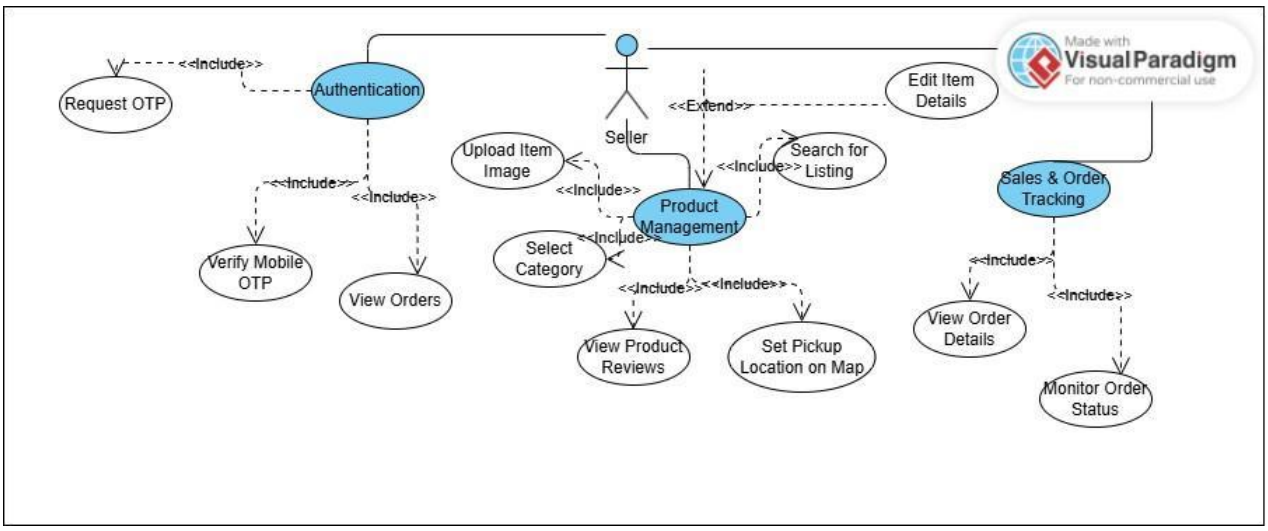


Figure 3: Studysphere Use Case Diagram (seller)

Use Case Diagram for Admin

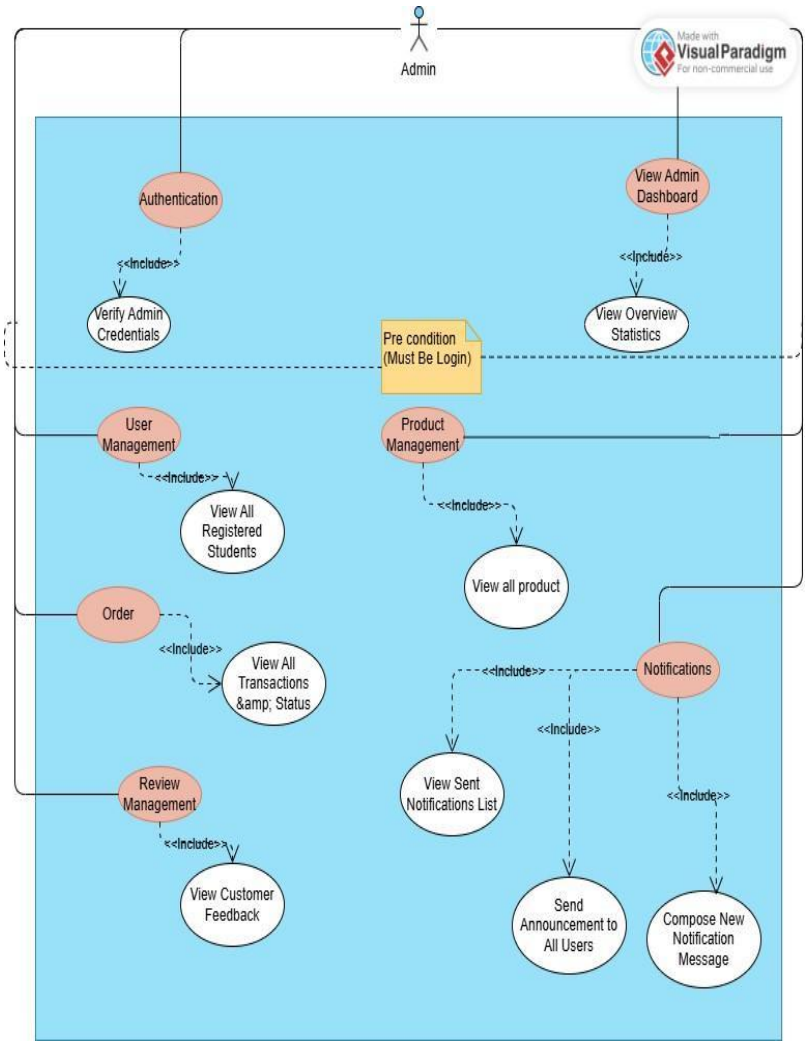


Figure 4: Studysphere Use Case Diagram (admin)

Studysphere Buyer Side App UI

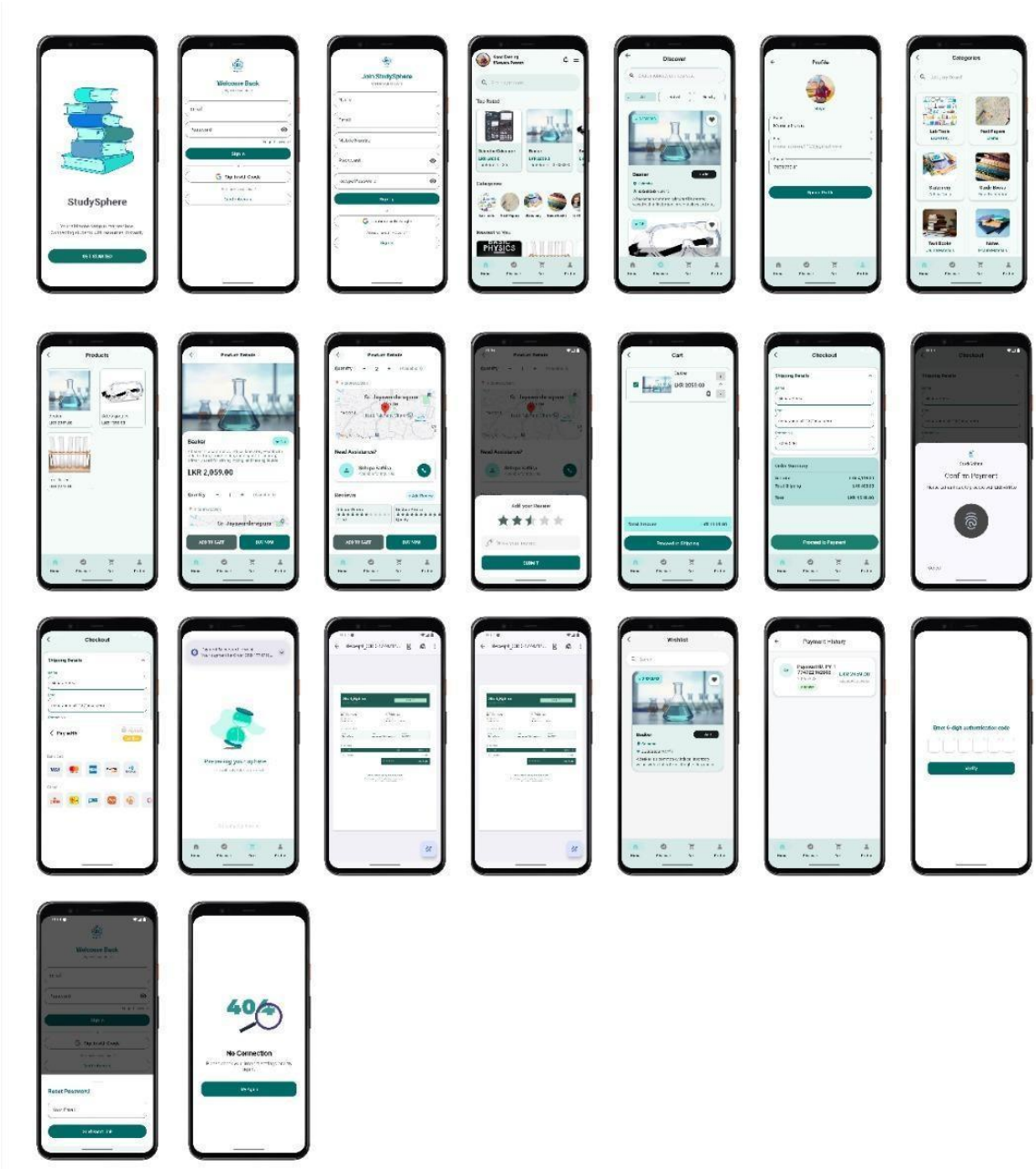


Figure 5: Studysphere Buyer Application UI

Studysphere Seller Side App UI

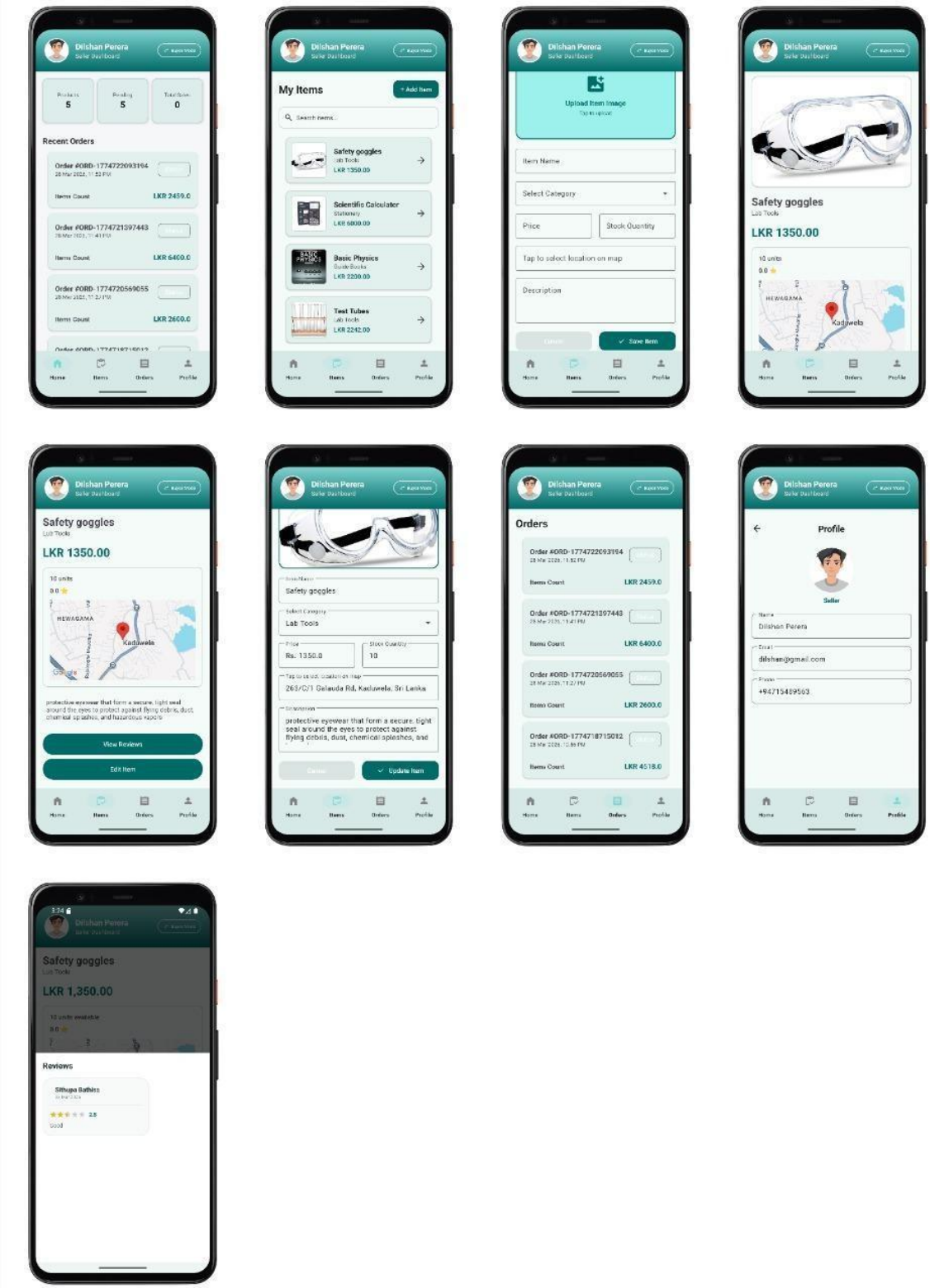


Figure 6: Studysphere Seller side Ui

Studysphere Admin Panel

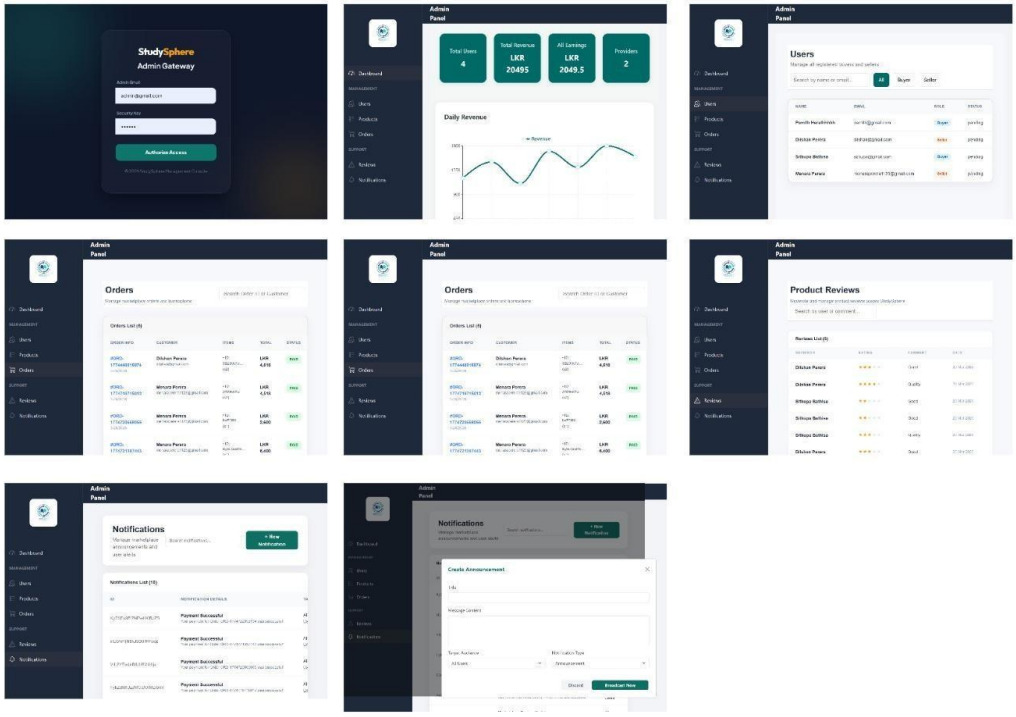


Figure 7: Studysphere Creator Admin Panel

5. Implementation

The implementation of StudySphere was strategically designed to address significant limitations found in existing academic resource platforms. One of the primary technical challenges was managing the efficient loading and filtering of vast product data. Traditional systems often suffer from high latency when loading categories and products. To solve this, I implemented an Asynchronous Data Fetching logic using Firebase Firestore snapshots, which ensures that categories and products load progressively without freezing the UI.

Another critical hurdle was the "Discovery" feature's accuracy. Most existing student marketplaces lack localized search, making it difficult to find items within a specific campus radius. I addressed this by integrating the Google Maps SDK and implementing a custom distance-calculation algorithm. This allowed the system to dynamically Sort and Filter products based on the buyer's real -time GPS coordinates and product ratings.

Security was another area where typical systems fail, often relying on simple passwords for high-value transactions. In StudySphere, I implemented a Multi-layered Security approach. For the role transition from Buyer to Seller, I solved the identity verification problem by integrating a Mobile OTP API, ensuring that only verified users can post listings. Furthermore, to secure the payment process, I integrated Biometric (Fingerprint) Authorization as a prerequisite for the PayHere gateway. These technical implementations not only resolved the performance bottlenecks of existing systems but also created a highly secure and location-aware marketplace tailored for students.

6. Results and Evaluation

The evaluation of StudySphere confirms that the primary goals of developing a secure, location-aware marketplace for students have been successfully achieved. A standout success of this implementation is the application’s Offline Resilience, which was specifically designed to handle connectivity issues. To ensure a seamless user experience, I integrated a local SQLite database to store and retrieve Payment Notifications, allowing users to view their transaction alerts even without an active internet connection. Furthermore, the system utilizes the device's Internal Storage to automatically save Payment Receipts as PDF files immediately after a transaction. Testing verified that even in a "No Internet" state, users can reliably access their critical transaction history and saved receipts.

During the development phase, several strategic refinements were made to the original proposal to optimize system efficiency. Product Management (Add, Edit, and Delete) was moved entirely from the Admin panel to the Seller module, empowering resource owners to manage their inventory in real-time. While advanced social features—such as "long-press to wishlist" and seller-specific trend analytics—were not prioritized in this phase to focus on the core security of Biometric Checkout and OTP Verification, the system remains highly stable and functional. The Admin Panel effectively facilitates community moderation by monitoring User Reviews and broadcasting Real -time Notifications. Overall, I am confident that StudySphere provides a robust and secure foundation that successfully meets the academic exchange needs of the student community.

7. Future Work

While the current version of StudySphere provides a stable foundation for academic exchange, several enhancements have been identified for future development to further enrich the user experience. The most critical planned feature is an In-App Real-Time Messaging System powered by Firebase Cloud Messaging (FCM). This would allow buyers and sellers to communicate directly within the app to negotiate prices and arrange pickups, eliminating the need for external contact methods. Additionally, I intend to implement advanced UI gestures, such as long-press actions to quickly manage wishlist items, and a more comprehensive Social Engagement module that includes detailed seller feedback and ratings. To support platform growth, future updates will also focus on Advanced Analytics for the Admin panel to track market trends and AI-driven content moderation to automatically flag inappropriate listings. These additions will ensure that StudySphere evolves into a fully interactive, automated, and highly scalable community hub for students.

8. Conclusion

In conclusion, StudySphere successfully addresses a significant gap in the academic ecosystem by providing a dedicated, secure, and location-aware marketplace for students. The project demonstrates how mobile technology can be leveraged to solve real-world problems, specifically the high cost and fragmentation of obtaining educational resources. By integrating Google Maps API for proximity-based searching, the application ensures that students can find essential materials like textbooks, lab equipment, and study notes within their immediate campus environment, thereby reducing logistics and costs.

9. Reflection of Learning

The StudySphere project provided an invaluable opportunity for double-loop learning, where the focus shifted from merely acquiring technical skills to critically examining the assumptions and decision-making processes that guided the development. Initially, my assumption was that a cloud-dependent architecture using Firebase would be sufficient for all system requirements. However, the reality of inconsistent campus network connectivity challenged this, forcing a revaluation of my "cloud-first" approach. This led to the strategic implementation of SQLite for local payment notifications and Internal Storage for PDF receipts, highlighting that offline resilience is just as critical as real-time connectivity in a real-world student environment.

Another key learning was the realization that user trust is built on security rather than just a wide array of features. While the initial plan included various social engagement tools, challenges in ensuring verified transactions led me to prioritize Biometric (Fingerprint)

Authorization and Mobile OTP Verification. This shift reinforced the importance of designing with a "security-first" mindset, especially when handling financial transactions within a peer-to-peer marketplace. Furthermore, the decision to move Product Management from the Admin to the Seller module reflected a deeper understanding of decentralized systems, ensuring data accuracy by giving resource owners direct control over their listings.

The project also underscored the value of modularity and organized architecture. By structuring the application into clear packages such as adapters, helpers, and models, I improved the system's maintainability and scalability. This iterative process of testing and refining assumptions such as prioritizing core security over minor UI gestures—has not only improved the application but also deepened my understanding of how to balance technical feasibility with user expectations. In conclusion, StudySphere was a journey of challenging my initial concepts and adapting to practical constraints, lessons that will undoubtedly shape my professional approach to future software engineering challenges.

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